Multispectral Object Detection via Cross-Modal Conflict-Aware Learning

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ABSTRACT

Multispectral object detection has gained significant attention due to its potential in all-weather applications, particularly those involving visible (RGB) and infrared (IR) images. Despite substantial advancements in this domain, current methodologies primarily rely on rudimentary accumulation operations to combine complementary information from disparate modalities, overlooking the semantic conflicts that arise from the intrinsic heterogeneity among modalities. To address this issue, we propose a novel learning network, the Cross-modal **C**onflict-**A**ware **L**earning Network (CALNet), that takes into account semantic conflicts and complementary information within multi-modal input. Our network comprises two pivotal modules: the Cross-Modal Conflict Rectification Module (CCR) and the Selected Cross-modal Fusion (SCF) Module. The CCR module mitigates modal heterogeneity by examining contextual information of analogous pixels, thus alleviating multi-modal information with semantic conflicts. Subsequently, semantically coherent information is supplied to the SCF module, which fuses multi-modal features by assessing intra-modal importance to select semantically rich features and mining inter-modal complementary information. To assess the effectiveness of our proposed method, we develop a two-stream one-stage detector based on CALNet for multispectral object detection. Comprehensive experimental outcomes demonstrate that our approach considerably outperforms existing methods in resolving the cross-modal semantic conflict issue and achieving state-of-the-art accuracy in detection results.

CCS CONCEPTS

• **Computer systems organization → Scene understanding**; **Object Detection**; **Computer vision problems**.

KEYWORDS

multispectral detection, multi-modal fusion, cross-modal learning

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1 INTRODUCTION

Object detection is a pivotal task in computer vision that involves the precise identification and localization of objects within an image and the assignment of distinct object classes to those objects [5, 6, 55, 58, 64]. However, traditional methods encounter limitations in detecting objects accurately under poor lighting conditions, where visible (RGB) images often contain sparse information [3]. To address this challenge, infrared sensors have emerged as a promising alternative for object detection in unlit conditions [26]. Unlike natural light images, infrared (IR) sensors can detect the physical intensity of infrared radiation and are less susceptible to lighting transformations [16].

To improve the accuracy of object detection in all-weather conditions, multispectral object detection methods leveraging the rich complementary information from RGB and IR images have emerged as a promising solution [20, 37]. These methods have exhibited enhanced detection accuracy and the capability to enable real-time object detection in all-weather scenarios [60]. Multispectral object detection has garnered significant attention due to its potential practical and societal benefits in applications such as automated driving [23, 57], surveillance [33], and crowd counting [35]. Recent research efforts have concentrated on devising novel cross-modal fusion techniques that effectively align and integrate features from two modalities to enhance the accuracy and effectiveness of multispectral object detection [3, 7, 18, 22, 59]. TSRA [48] also developed

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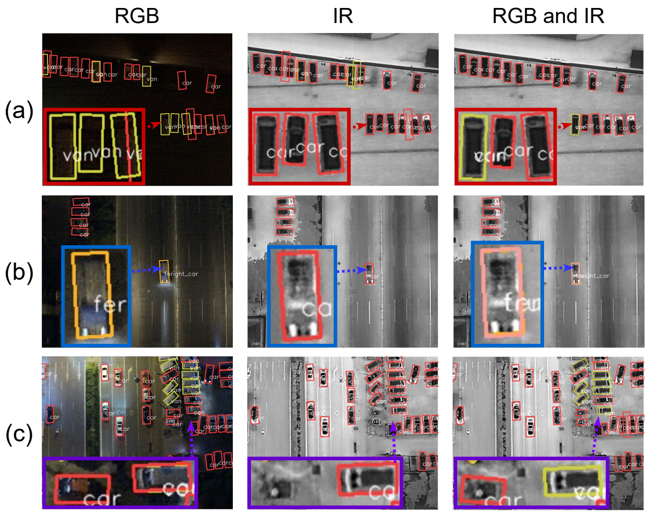

Figure 1: Illustrations of the information with semantic conflicts in multispectral object detection for different inputs. RGB and IR inputs fuse using feature concatenation. The different color bounding box is the area of interest. (a) Conflicting detection can emerge from inconsistent information regarding detected objects. (b) Conflicting classification of RGB and IR images can confuse classification. (c) Conflicting background information between RGB and IR images can lead to redundancy and false detection.

an alignment framework that addresses the challenge of weakly misaligned multispectral images to augment the accuracy of multispectral object detection.

However, the inevitable heterogeneity of multispectral images obtained from different sensors can lead to conflicts in the complementary information of multi-modal data, which can hinder the accuracy of object classification and localization [15]. Semantic conflicts arise when two or more modalities perceive the same object information but assign inconsistent semantic categories to the object. In such cases, fine-grained information crucial for classification may be overshadowed during the fusion process, leading to confusion of object categories and background. As illustrated in Fig. 1, cross-modal semantic conflict leads to object classification confusion and mislocalization, which impacts the final detection and classification results. Unfortunately, the aforementioned feature fusion-based methods ignore the semantic conflict problem in the cross-modal fusion process, reducing the discriminability of cross-modal fusion features and leading to a confusing representation of cross-modal semantic information.

To address this challenge, we propose a novel method named the Conflict-Aware Cross-Modal Learning Network (CALNet), comprising two core modules: the Cross-Modal Conflict Rectification (CCR) module and the Selected Cross-Modal Fusion (SCF) module. The CCR module operates by projecting multispectral features into a shared joint space, establishing nearest neighboring patch relationships, and extracting contextual information associations between patterns. Next, a transformer is employed to aggregate a single-modal patch with some of the most similar patches from outside that modal, reducing the heterogeneity of between patches between corresponding locations using extra-modal contextual information, thus rectifying cross-modal semantic conflicts.

Moreover, existing methods often employ simple fusion strategies that measure the importance of single-modal features to achieve fusion [19, 36, 42, 48, 59]. These methods can lead to valid complementary information being overwhelmed by a large amount of redundant background information (e.g., nighttime RGB images with partial illumination.). To overcome this issue, we introduce the Selected Cross-Modal Fusion (SCF) module, which selects significant features for fusion. The SCF module first separately calculates the importance of intra-modal feature weights and selects the significant features. On the other hand, the inter-modal features are aggregated using spatial and channel attention methods. Finally, the inter-modal features are reversed into the intra-modal features, so that the unimodal features incorporate the multi-modal consistent representation. This module enhances the discriminative properties of cross-modal fusion features, improves cross-modal fusion by selecting rich semantic features, and leads to better performance in multispectral object detection.

Contributions: The main contributions of this paper are:

- We propose a CALNet for multispectral object detection, comprising CCR and SCF modules. Our CCR module diminishes inter-modal heterogeneity by mining contextual information of analogous pixels to rectify conflicting information in multispectral features. After that, our SCF module selects rich semantic features by ranking the importance of intra-modal features and aggregates inter-modal features.
- We devise a one-stage multispectral oriented object detector employing CALNet and conduct extensive experiments to demonstrate the effectiveness of our method in addressing the semantic conflict issue in cross-modal fusion.
- Comprehensive experiments on two multispectral object detection benchmarks: DroneVehicle and LLVIP. Our proposed method achieves state-of-the-art performance on both datasets, achieving higher accuracy and speed.

2 RELATED WORK

Oriented Object Detection: Convolutional Neural Networks (CNNs) have emerged as a critical tool in advancing object detection in aerial imagery, particularly in the RGB domain [2, 4, 8, 43]. In recent years, deep learning-based object detection methods have achieved state-of-the-art performance [1, 9, 12, 49]. To further the development of aerial object detection, Xie et al.[41] introduced the DOTA dataset. The aforementioned methods have achieved promising results in the field of object detection based on RGB. Despite the success of these methods, object detection based on RGB images may struggle to achieve satisfactory segmentation results in scenes with insufficient illumination [28]. Consequently, researchers are exploring the use of multispectral data to improve the accuracy and robustness of object detection [29, 53, 54, 61]. To address this challenge and ensure high performance, we propose a two-stream oriented object detector based on one-stage.

Multispectral Object Detection: Prior object detection methods have primarily utilized RGB input images, which are insufficient for capturing crucial information in low-light conditions [30–32, 62].

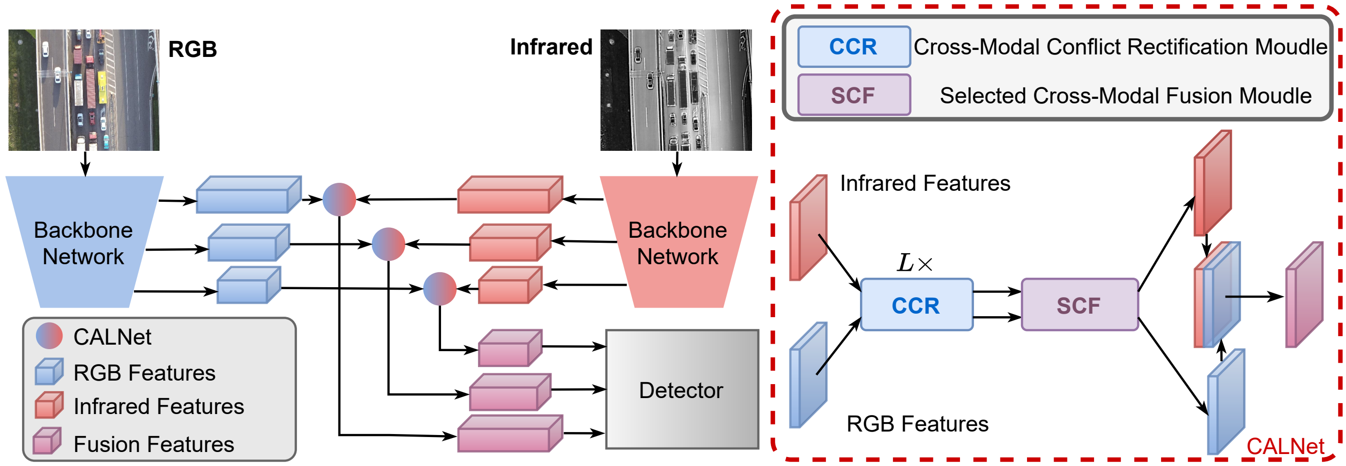

Figure 2: The overall architecture of the proposed multispectral object detection framework. It comprises a two-stream backbone network taking RGB and IR inputs, a novel cross-modal conflict-aware learning network (CALNet), and a one-stage oriented detector for classification and localization. CALNet comprises a cross-modal conflict rectification (CCR) module and a selected cross-modal fusion (SCF) module, which captures the heterogeneous and complementary information across the two modalities.

As a result, RGB-based object detection techniques are generally not robust to variations in illumination [14, 50, 56]. In recent years, multispectral object detection has emerged as a vital application for self-driving cars and video surveillance due to its ability to offer robustness to illumination variations [19]. Multispectral object detection involves exploiting cross-modal complementary information to enhance the accuracy of object detection, and several existing works have achieved progress in multi-modal fusion [38, 45].

To address the issue of weak misalignment in multispectral data, the AR-CNN [52] proposes a Region-based Alignment (RFA) module that predicts the movement changes of regions between two modalities. TSRA [48] introduces an alignment framework to mitigate the weak misalignment problem of modal images. MBNet [59] proposes a modal perception fusion method that measures the importance of RGB and IR information in object detection. Similarly, Kim et al.[18] introduced detectors that can seamlessly receive both visible and infrared information, thus enabling more accurate and robust object detection in real-world scenarios.

Nonetheless, the aforementioned methods focus on exploiting the complementary information of multi-modal data during the fusion of cross-modal information, without accounting for the issue of semantic conflict. To address this problem, this study proposes CALNet, which fully considers both semantic conflict and complementary information in multispectral data and effectively fuses them to enhance object detection performance.

3 METHODOLOGY AND ANALYSIS

Overall Architecture. Fig.2 displays the comprehensive architecture (CALNet) of the cross-modal conflict-aware learning Network framework, a proposed multispectral object detector for aerial object detection. The framework comprises a two-stream backbone network and an oriented detection head. For simplicity, we omit the details of the oriented detection head in the feature pyramid network [24] description. Specifically, the first stream uses a backbone to extract multi-level RGB features, while the second stream extracts multi-level infrared features. To address the challenge of semantically conflicting information in multispectral data, we introduce a CCR module that rectifies heterogeneity in high-level RGB and infrared features by employing similar pixel contextual information. Furthermore, we develop an SCF module that selects significant features for cross-modal fusion, enhancing the discriminative properties of the fused features. Finally, we input the cross-modal fused features from the last three layers of features into the oriented detection head to predict the final object detection results, achieving both optimal performance and efficient computation.

Consistent with numerous multispectral object detection models, our proposed multispectral object detector incorporates a two-stream network, composed of two symmetrical unimodal feature extractors. These extractors have identical network structures but distinct parameters, enabling the extraction of RGB and infrared features at various levels. A two-stream backbone is employed to extract RGB features $\{\boldsymbol{r}_t \mid t = 1, 2, \cdots, 5\}$ and IR features $\{\boldsymbol{i}_t \mid t = 1, 2, \cdots, 5\}$ at spatial sizes of 1/2, 1/4, $\cdots$, 1/32 of the input data. Following this, the last three layers of features are selected for input into our CALNet. The following equation describes the process:

$$\begin{bmatrix} \widetilde{\boldsymbol{r}_t} \\ \widetilde{\boldsymbol{i}_t} \end{bmatrix} = \mathrm{SCF}\left(\mathrm{CCR}\left(\begin{bmatrix} \boldsymbol{r}_t \\ \boldsymbol{i}_t \end{bmatrix}\right)\right). \tag{1}$$

Finally, we add features $\widetilde{\boldsymbol{r}_t}$ and $\widetilde{\boldsymbol{i}_t}$ as follows: $\mathcal{G} = \mathrm{Add}\left(\widetilde{\boldsymbol{r}_t}, \widetilde{\boldsymbol{i}_t}\right)$. The output feature $\mathcal{G}$ is fed to the detector to predict the object confidence scores and bounding box coordinates. The design of the FPN is based on the work of Lin et al. [24], while the detector design refers to one-stage object detection methods[10, 21, 63]. Details of these modules are discussed in the subsequent sections.

3.1 Cross-Modal Conflict Rectification Module

Existing research primarily focuses on utilizing complementary information in multispectral data, yet it inadequately addresses

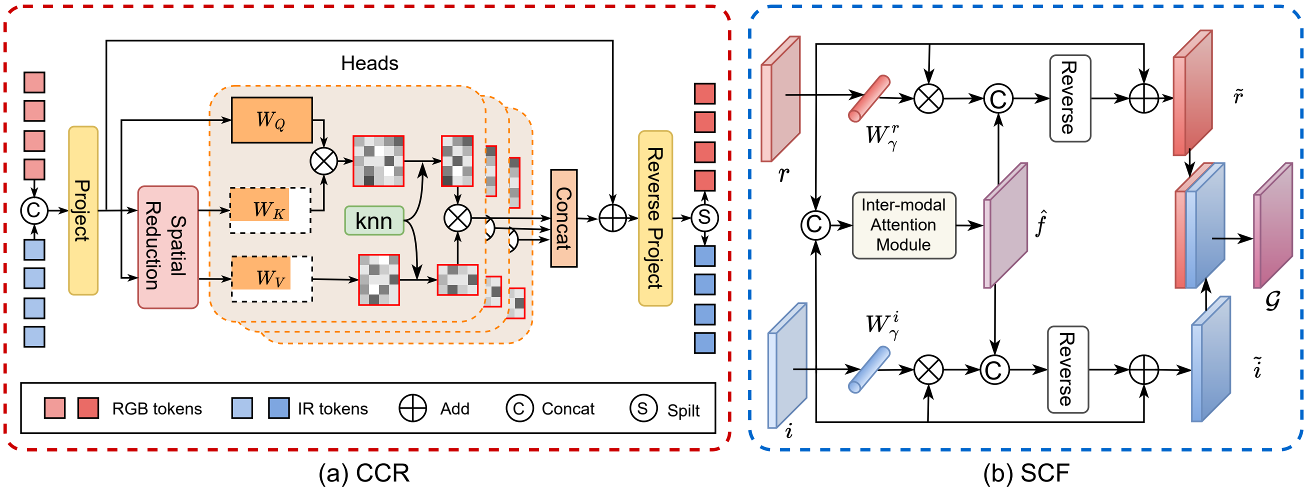

Figure 3: Structures of our CCR and SCF modules. (a) The structure of the CCR module. By selecting the cross-modal pixels with the highest similarity for interaction through KNN, the heterogeneity between multi-modal is reduced, thus alleviating the semantic conflict problem. (b) The structures of the SCF module.

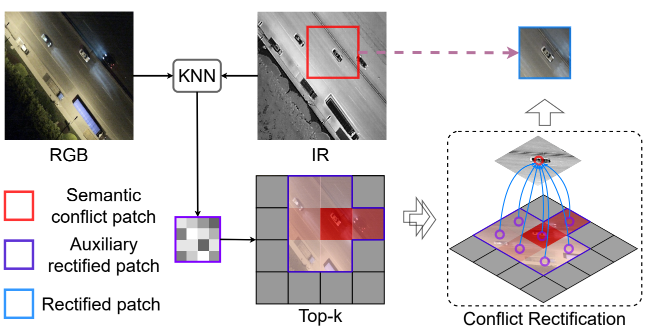

Figure 4: Illustration of our proposed rectified semantic conflicts method. By measuring the similarity between pairs of modalities in the same space, the most relevant contextual information outside the modalities is filtered out to assist in rectifying the semantic conflicts between the modalities.

the semantic conflict arising from multi-modal. This shortcoming hampers the impact of fine-grained semantic information on object class inference. To tackle this issue, we introduce the CCR module. The semantic conflict is resolved by employing the transformer's dynamic modeling capabilities to guide contextual information exchange between modalities, consequently mitigating cross-modal semantic conflicts and reducing modal heterogeneity. Fig. 3(a) depicts the structure of the CCR module. Before implementing the CCR module, we acquire features $\boldsymbol{r}$ and $\boldsymbol{i}$ from the two-stream network and join them to compute pixel similarity: $\boldsymbol{f} = \text{Concat}(\boldsymbol{r}, \boldsymbol{i})$.

We first identify the nearest neighboring patch relationship within this space to extract the contextual information association between patterns. Since features from distinct spaces cannot directly compute similarity, we project these features from the disparate spaces into a shared joint space. Specifically, the multispectral feature $\boldsymbol{f}$ is projected through a linear layer, forming a query $W_Q \in \mathbb{R}^{n\times d_q}$, a key $W_K \in \mathbb{R}^{n\times d_k}$, and a value $W_V \in \mathbb{R}^{n\times d_v}$, where $n = m \times h \times w$ represents the number of patches and m denotes the number of multispectral inputs.

Given the high computational burden of the transformers, which is linearly associated with the number of multispectral inputs, we use a spatial reduction attention [39] to diminish the dimensionality of the *key* and *value*, thus enhancing the efficiency of the cross-modal conflict resolution. We employ an $s \times s$ depth-wise convolution with stride s to decrease the spatial size of W_K and W_V before the attention. The spatial reduction attention is defined as:

$$\begin{cases} W_K{}' = \text{DWConv}(W_K) \in \mathbb{R}^{\frac{n}{s^2}\times d_k}, \\ W_V{}' = \text{DWConv}(W_V) \in \mathbb{R}^{\frac{n}{s^2}\times d_k}, \end{cases} \tag{2}$$

where $\text{DWConv}(\cdot)$ denotes the depth-wise separable convolution. Each row of the query represents the features of each pixel point.

The CCR module comprises L layers, with sequences of the $(l+1)_{th}$ layer being $l = 0, ..., L-1$. The forward process of the exemplar feature in one layer can be expressed as:

$$\begin{cases} f^l = f^l + \text{Att}_{knn}\left(\text{LN}\left(f^l\right)\right), \\ f^{l+1} = f^l + \text{Reverse}\left(\text{LN}\left(f^l\right)\right), \end{cases} \tag{3}$$

where LN represents LayerNorm, and Reverse denotes the feed-forward network, serving to reverse project back into the RGB and IR modalities. The key to our design is attention.

As illustration in Fig. 4, for the i-th query patch, we first calculate the euclidean distance of all keys, then obtain its k-nearest neighbors [11] from the keys and values, and ultimately compute the scaled dot product attention. Next, a similarity attention is employed to aggregate a unique model patches with some of the most similar patches outside of that model. Reduce the heterogeneity between patches by using extra-modal contextual information to auxiliary rectify semantic conflicts. The similarity attention is calculated using the following formula:

$$\boldsymbol{A}_i = \operatorname{softmax}\left(\frac{\left\langle W_{Q_i}, \left(W_{K_{j_1}}, \ldots, W_{K_{j_k}}\right)\right\rangle}{\sqrt{d}}\right), W_{K_{j_k}} \in \mathcal{N}_i^k, \quad (4)$$

where $1/\sqrt{d}$ is the scaling factor. Since the computation of the Euclidean distance for all keys against each query is slow, we take advantage of the matrix multiplication to speed it up. As with the vanilla attention, all queries and keys are computed by dot product:

$$\boldsymbol{A} = \boldsymbol{f}^{\boldsymbol{l}} W_Q \cdot \boldsymbol{f}^{\boldsymbol{l}} W_K{}^{\top}, \quad (5)$$

and then row-wise top-k elements are selected for softmax computing. The formula attention is:

$$\begin{gathered} [\boldsymbol{X}]_{ij} = \begin{cases} 1 & , \boldsymbol{A}_{ij} \in \text{ top-k}\,(\text{row} j) \\ -\infty & , \text{ otherwise} \end{cases}, \\ \text{Att}_{knn}\left(\boldsymbol{f}^{\boldsymbol{l}}\right) = \operatorname{softmax}\left(\frac{\boldsymbol{A} \cdot \boldsymbol{X}}{\sqrt{d}}\right) \cdot \boldsymbol{f}^{\boldsymbol{l}} W_V, \end{gathered} \quad (6)$$

where $\boldsymbol{X}$ is an adjacency matrix based on top-k elements. The resulting matrix is the value shape $\text{Att}_{knn}(\boldsymbol{f}^{\boldsymbol{l}}) \in \mathbb{R}^{n\times k}$, with k denoting the k most similar nodes calculated based on the metric distance. Following this, the feature can be passed through subsequent rounds of the CCR module to further reduce heterogeneity between cross-modal. Semantic conflicting information is rectified by cross-modal contextual information, thus reducing the heterogeneity of cross-modal images and producing semantically consistent information.

3.2 Selected Cross-Modal Fusion Module

Prior research has employed a simple parameter to assess the importance of RGB and IR modalities. However, low information value regions in unimodal data can affect the scoring of high information value regions (e.g., RGB images with partial lighting). To address this issue, we propose a novel Selected Cross-Modal Fusion (SCF) module to capture rich semantic features, as depicted in Fig. 3 (b). This module filters out low-semantic features by adaptively assigning weights to intra-modal and incorporate inter-modal consistent information to enrich the intra-modal semantic representation.

We first obtain the weight coefficients γ for each channel by normalizing features derived from the intra-modal. Next, we multiply each channel with the corresponding weight coefficients to attain a sparse weight penalty, which suppresses less significant channel expressions. Given modal $\boldsymbol{r}$ and $\boldsymbol{i}$, the self-adaptive weight of intra-modal features is defined as:

$$\boldsymbol{r} = \mathcal{S}\left(W_\gamma^r \cdot \text{BN}\,(\boldsymbol{r})\right), \boldsymbol{i} = \mathcal{S}\left(W_\gamma^i \cdot \text{BN}\,(\boldsymbol{i})\right) \quad . \quad (7)$$

where the weights are obtained as $W_\gamma = \gamma_i / \sum_{j=0} \gamma_j$. BN is batch normalization. $\mathcal{S}$ is the sigmoid function.

By assigning weights to the selection, we filter out numerous insignificant features under inadequate lighting conditions. In this manner, we draw refined intra-modal features. To incorporate inter-modal features, we concatenate these features to obtain $\boldsymbol{f}$. Inspired by [40], we fuse RGB and IR features to further reduce inter-modal heterogeneity using the inter-modal attention module. The detailed inter-modal attention module is as follows:

$$\mathcal{H}_c(\boldsymbol{f}) = \mathcal{S}(\text{MLP}(\text{AvgPool}(\boldsymbol{f})) + \text{MLP}(\text{MaxPool}(\boldsymbol{f}))), \quad (8)$$

where $\mathcal{S}$ is the sigmoid function, and $AvgPool$ and $MaxPool$ signify cross-modal global pooling operations along the d, h, and w dimensions. The refined feature $\boldsymbol{f_c} \in \mathbb{R}^{m\times h\times w\times d}$, derived from the inter-modal attention, can be expressed as $\boldsymbol{f}c = \mathcal{H}c \cdot \boldsymbol{f}$. On the other hand, the spatial attention map $\mathcal{H}_s \in \mathbb{R}^{h\times w\times d}$ undergoes further multiplication with $\boldsymbol{f_c}$, compressing the inter-modal dimension. This process results in injected inter-modal features with the same dimensions as the intra-modal features, as follows:

$$\mathcal{H}_s(\boldsymbol{f_c}) = \mathcal{S}(C([\text{AvgPool}(\boldsymbol{f_c}); \text{MaxPool}(\boldsymbol{f_c})]), \quad (9)$$

where C represents a convolutional filter, and $\widehat{\boldsymbol{f}} = \mathcal{H}s$. Next, we reverse the inter-modal features to the intra-modal features as:

$$\widetilde{\boldsymbol{r}} = \boldsymbol{r} + \text{R}_{rgb}(\boldsymbol{r}, \widehat{f}), \widetilde{\boldsymbol{i}} = \boldsymbol{i} + \text{R}_{ir}(\boldsymbol{i}, \widehat{f}) \quad , \quad (10)$$

where $\text{R}(\cdot)$ denotes the reverse matrix which employs depth-wise separable convolution. Finally, we obtain RGB and IR features with consistent representations, which are combined to produce the final features. By selecting features to obtain rich semantic intra-modal features while interacting with features, the consistency of inter-modal features is further augmented during the fusion.

Detection Head: The DroneVehicle annotation box encompasses directional information. To tackle the angle regression problem, we draw inspiration from [46] and transform it into a 180-degree classification problem. This approach enables learning of the object orientation by converting the angle of the object into a classification problem within the 0 to 180-degree range.

4 EXPERIMENTAL RESULTS

4.1 Dataset and Evaluation Metrics

DroneVehicle [36] comprises aerial RGB-IR images captured by drones, encompassing various scenes from an aerial perspective. It contains five categories of target objects. The dataset comprises 28,439 pairs. The images are classified into three distinct scenes: day, night, and dark night.

LLVIP [17] is a recently released large-scale RGB-IR pedestrian dataset designed for low-light scenes. Captured by surveillance cameras at 26 street locations, the majority of images were taken in low or no light conditions, and all images are strictly aligned. The dataset comprises 33,672 images, comprising 16,836 image pairs.

Evaluation metrics. We employ the COCO-style metric of mAP to assess the accuracy of multispectral object detection. For mAP calculations, an intersection over union threshold is utilized to determine true and false positives.

Implementation Details. We implemented the proposed network using PyTorch 1.10.1 and CUDA 11.3, and trained our method on an NVIDIA RTX 3090. During training, we employed a two-stream feature extraction architecture initialized with pre-trained weights in the backbone network, as illustrated in Fig.2. The input image size is set to 640×640. To enhance detection performance, we employed a data mosaic as a data augmentation technique during training. We followed the module configuration setup in pvt [39] for our experiments. The stochastic gradient descent (SGD) algorithm was used for optimization with a momentum of 0.850 and a weight decay of 0.001. We used a batch size of 4, and the network was trained for 50 epochs with a learning rate of 0.035 to achieve optimal performance.

Table 1: Comprehensive experiments on the DroneVicle dataset. We compare the CALNet method with single-modal and multispectral detectors. For the fairness of the experiment, The two-stream backbone network of the multispectral detector is set ResNet50 and DarkNet53. The OBB indicates oriented bounding box.

DroneVehicle val.									
Detectors	Backbone	Type	Car	truck	freight car	bus	van	mAP@0.5	Modal
RetinaNet(OBB)[25]	ResNet50	one-stage	78.45	34.39	24.14	69.75	28.82	47.11	RGB
Faster R-CNN(OBB)[34]	ResNet50	two-stage	79.69	41.99	33.99	76.94	37.68	54.06	
RoITransformer[8]	ResNet50	two-stage	61.55	55.05	42.26	85.48	44.84	61.55	
S2ANet[12]	ResNet50	one-stage	79.86	50.02	36.21	82.77	37.52	57.28	
Oriented R-CNN[43]	ResNet50	two-stage	80.26	55.39	42.12	86.84	46.92	62.30	
YOLOv5(OBB)[17]	DarkNet53	one-stage	89.15	59.72	42.95	78.75	43.88	62.89	
RetinaNet(OBB)[25]	ResNet50	one-stage	88.81	35.43	39.47	76.45	32.12	54.45	IR
Faster-R-CNN(OBB)[34]	ResNet50	two-stage	89.68	40.95	43.10	86.32	41.21	60.27	
RoITransformer[8]	ResNet50	two-stage	89.64	50.98	53.42	88.86	44.47	65.47	
S2ANet[12]	ResNet50	one-stage	89.71	51.03	50.27	88.97	44.03	64.80	
Oriented R-CNN[43]	ResNet50	two-stage	89.63	53.92	53.86	89.15	40.95	65.50	
YOLOv5(OBB)[17]	DarkNet53	one-stage	89.29	59.89	49.51	86.72	45.06	65.49	
Halfway Fusion (OBB) [27]	ResNet50	two-stage	89.85	60.34	55.51	88.97	46.28	68.19	RGB+IR
CIAN(OBB) [51]	ResNet50	one-stage	89.98	62.47	60.22	88.9	49.59	70.23	
AR-CNN (OBB) [52]	ResNet50	two-stage	90.08	64.82	62.12	89.38	51.51	71.58	
TSFADet [48]	ResNet50	two-stage	89.88	67.87	63.74	**89.81**	53.99	73.06	
Cacade-TSFADet [48]	ResNet50	two-stage	90.01	69.15	65.45	**89.70**	55.19	73.90	
CALNet	ResNet50	one-stage	90.11	70.66	67.48	89.74	55.01	74.60	
CALNet (Ours)	DarkNet53	one-stage	**90.27**	**73.69**	**68.67**	89.70	**59.74**	**76.41 (↑3.35)**	

DroneVehicle test.									
Detectors	Backbone	Type	car	truck	freight car	bus	van	mAP@0.5	Modal
RetinaNet (OBB) [25]	ResNet50	one-stage	67.50	28.24	13.72	62.05	19.26	38.10	RGB
Faster R-CNN (OBB) [34]	ResNet50	two-stage	67.88	38.59	26.31	66.98	23.2	44.59	
RoITransformer [8]	ResNet50	two-stage	68.13	44.17	29.08	70.55	27.64	47.91	
ReDet [13]	ResNet50	two-stage	69.48	47.87	31.46	77.37	29.03	51.04	
Gliding Vertex [44]	ResNet50	two-stage	75.77	46.08	33.75	68.05	38.72	52.48	
YOLOv5 (OBB) [17]	DarkNet53	one-stage	76.24	48.91	35.47	68.91	40.37	53.98	
RetinaNet (OBB) [25]	ResNet50	one-stage	79.86	32.84	28.05	67.32	16.44	44.90	IR
Faster R-CNN (OBB) [34]	ResNet50	two-stage	88.63	42.51	35.16	77.92	28.52	54.55	
RoITransformer[8]	ResNet50	two-stage	88.85	51.53	41.49	79.48	34.39	59.15	
ReDet [13]	ResNet50	two-stage	89.47	53.95	42.82	79.89	36.56	60.54	
Gliding Vertex [44]	ResNet50	two-stage	89.15	59.72	42.95	78.75	43.88	62.89	
YOLOv5 (OBB) [17]	DarkNet53	one-stage	89.29	59.89	49.50	86.72	42.06	65.49	
UA-CMDet [36]	ResNet50	two-stage	87.51	60.70	46.80	87.08	37.95	64.01	RGB+IR
TSFADet [48]	ResNet50	two-stage	89.23	72.00	54.15	88.08	48.77	70.44	
CALNet	ResNet50	one-stage	90.23	73.83	60.92	88.74	51.68	73.08	
CALNet (Ours)	DarkNet53	one-stage	**90.30**	**76.15**	**62.97**	**89.11**	**58.46**	**75.39 (↑4.95)**	

4.2 Comparisons with State-of-the-Art Methods

Comparsion on DroneVehicle test. We compared our method with competing methods on the test set of the DroneVehicle (shown in Table 1). Since there is a lack of cross-modal oriented detection methods for aerial scenes in existing studies, we evaluated our method and eight other state-of-the-art methods, with six single-modal and two multi-modal methods. Among the multi-modal methods, TSFDet achieved the highest detection accuracy (70.44% mAP@0.5). And instead, our proposed detector achieved 75.39% mAP@0.5, outperforming other methods and improving by 4.95% mAP@0.5 relative to the state-of-the-art method. Moreover, our method outperforms existing methods in all categories of the DroneVehicle dataset in terms of accuracy. Specifically, our method improves the accuracy of cars, trucks, freight cars, buses and vans by 1.04%, 4.15%, 8.82%, 1.03% and 9.69% mAP@0.5, respectively, further demonstrating the utility of our proposed method for alleviating the confusion classification problem caused by semantic conflicts.

Comparsion on DroneVehicle val. As shown in Table 1, we evaluated our method against eleven other advanced methods, including five multi-modal methods. The same methods using RGB and IR images outperformed the single-modal methods. Among

Table 2: Comprehensive experiments on the LLVIP dataset. All methods use RGB+IR as input.

Detectors	Backbone	Type	mAP
Faster RCNN [34]	ResNet50	two-stage	56.9
Cascade RCNN	ResNet50	two-stage	59.6
DCMNet [42]	ResNet50	two-stage	61.5
RetinaNet [25]	ResNet50	one-stage	56.9
Reppoints [47]	ResNet50	one-stage	57.6
CSSA [47]	ResNet50	one-stage	59.2
ImageBind [47]	ResNet50	one-stage	63.4
CALNet	ResNet50	one-stage	**63.4**
CALNet (Ours)	DarkNet53	one-stage	**63.9**

Table 3: Ablation experiments of CCR and SCF module on the DroneVehicle dataset. The SRA is spatial reduction attention. The baseline employed in this research is a one-stage two-stream object detector without CCR and SCF module that integrates two modalities through an additive fusion process.

Method	SRA	CCR	SCF	mAP@0.5
Baseline	–	–	–	68.15
Baseline + SCF	–	–	✔	71.92
Baseline + SRA	✔	–	–	72.46
Baseline + CCR	–	✔	–	74.13
Baseline + SRA + SCF	✔	–	✔	73.61
CALNet	–	✔	✔	**75.39**

Table 4: Performance of CALNet with the different number of CCR modules. s indicates the spatial reduction rate.

Method	Number of CCR	s	mAP@0.5
CALNet	(2,2,2)	(4,2,1)	72.91
	(3,3,3)	(4,2,1)	74.15
	(3,6,3)	(4,2,1)	**75.39**

the multi-modal methods, Cascade-TSFDet achieved advanced detection accuracy (73.90% mAP@0.5). In comparison, our CALNet achieves remarkable performance, with our proposed detector achieving 76.41% mAP@0.5, which is superior to other methods and improves by 3,35% mAP@0.5 relative to the state-of-the-art method.

Comparsion on LLVIP Dataset. LLVIP is a newly constructed large-scale RGB-IR dataset for low-light vision. We use the LLVIP dataset to evaluate our proposed method and compare it with state-of-the-art methods (shown in Table 2). Our CALNet achieved remarkable detection performance on this dataset, achieving the best results among all compared methods and outperforming them by 2.4% mAP. These experimental results demonstrate the advantages of our proposed method on the newly constructed dataset, thus demonstrating the effectiveness and robustness of CALNet.

Comparsion on Speed versus Accuracy. We compared the speed and accuracy of various detectors on an NVIDIA GV100 GPU. All detectors were tested at a batch size of 1. As shown in Fig. 5, our

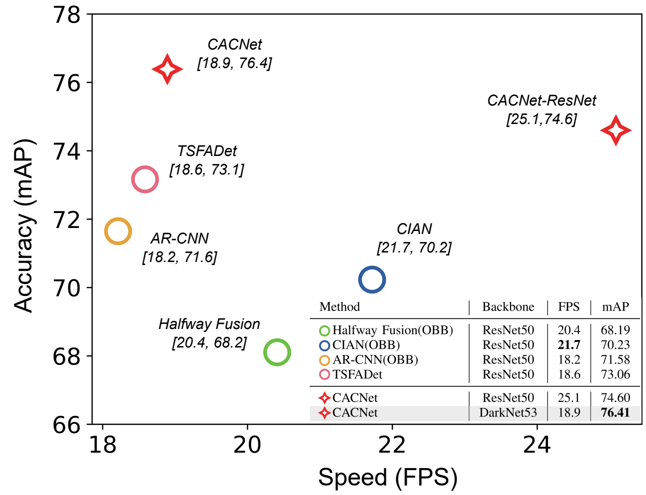

Figure 5: Speed vs. accuracy on the DroneVehicle using a GV100 GPU batch size of 1. Speed comparison using FPS.

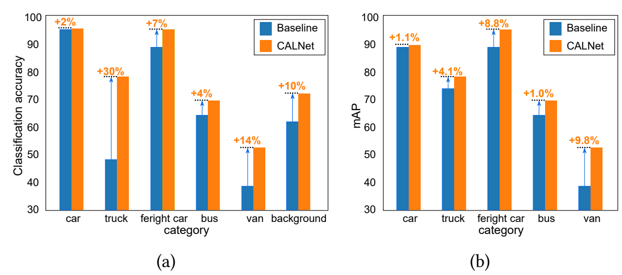

Figure 6: Comparison of classification accuracy on the DroneVehicle val with the baseline for each category. The baseline is a two-stream object detector fused by concatenation.

detector exhibits superior detection accuracy (75.39% mAP@0.5) compared to other multispectral detectors while maintaining a comparable speed (18.9 FPS).

4.3 Ablation Studies

We conduct ablation experiments on the DroneVehicle test set. Our baseline is a two-stream one-stage detector fused by concatenation.

Cross-Modal Conflict-Aware Module. The CCR module is designed to address the cross-modal semantic conflict issue by reducing inter-modal heterogeneity as given in Table 3. The baseline augmented with the CCR module exhibits a 1.67% mAP@0.5 improvement over the baseline with SRA. Additionally, CALNet outperforms CALNet with SRA by 1.78% mAP@0.5, attributable to the limited capacity of SRA to learn analogous pixel representations across modalities. The SCF module remains stable in terms of enhancement for both SRA and CCR modules, independent of cross-modal similarity attention. Compared to the baseline, the CCR module alone contributes to a 4.31% mAP@0.5 increase. Furthermore, Table 4 demonstrates the effectiveness of CALNet using different numbers of CCR modules.

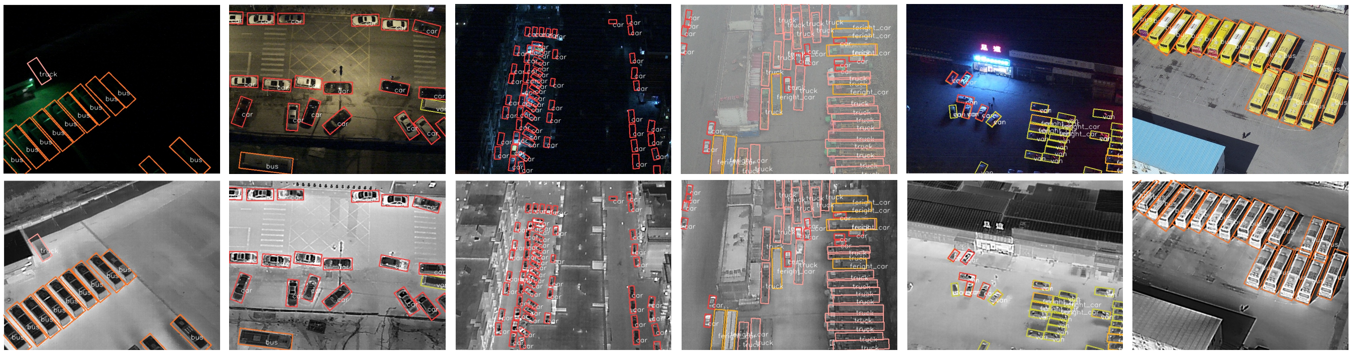

Figure 7: Visualization results of CALNet on the DroneVehicle, using different color boxes for different categories. The performance is excellent in locating tiny objects, dense objects, and fine-grained classification under various light conditions.

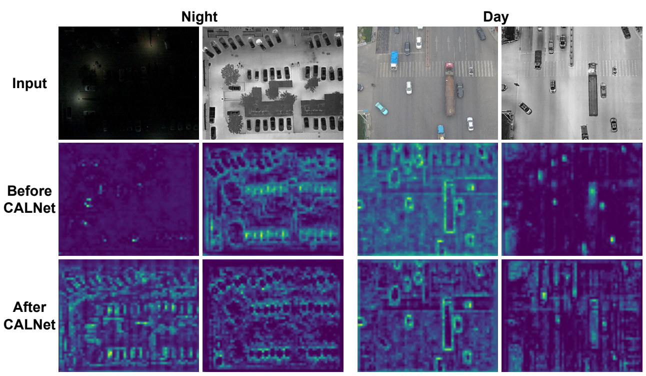

Figure 8: Visualization of the features in Stage3 (r_3 and i_3) before and after the CALNet. The two modal feature maps are compensated using complementary information.

Selected Cross-Modal Fusion Module. Under identical experimental conditions, we execute experiments to validate the effectiveness of SCF. The aim of SCF is to adaptively select features with rich semantics by evaluating feature importance. As depicted in Table 3, the baseline+SCF yields a 3.77% mAP@0.5 enhancement to the model, while SCF continues to provide a 1.2% mAP@0.5 improvement to the model featuring the CCR module. Consequently, the CCR and SCF modules cooperate to further advance cross-modal information fusion. We observe that CCR alone outperforms SCF module, indicating that cross-modal fusion favors the resolution of semantically conflicting information. The amalgamated experimental results demonstrate that integrating the advantages of each component improves the outcomes of DroneVehicle and LLVIP.

4.4 Visualization Results

Classification Performanc Conflicting information between different modalities can create semantic ambiguity and significantly affect the categorization of objects. To investigate the performance of CALNet in addressing cross-modal semantic conflicts, we compared the classification accuracy of CALNet and the baseline model on the DroneVehicle dataset (shown in Fig. 6(b)). This dataset consists of five classes of automobiles, presenting a significant challenge for fine-grained classification due to the substantial ambiguity among classes. The fine-grained classification further compound the classification confusion problem. Fig. 6(a) illustrates that the baseline model has difficulty classifying trucks and vans. In contrast, CALNet improves the classification accuracy of trucks and vans by 30% and 10%, respectively, and by 11.4% on average for each category.

Feature Visualization: Fig. 8 presents the visualization results of CALNet. The RGB and IR modalities exhibit limitations in accurately capturing both object and background features due to their distinct characteristics. Following the CALNet process, object features are emphasized, while background features are globally integrated. This integration results in the extraction of meaningful background information and the reduction of background noise. The proposed CALNet effectively combines cross-modal features, accelerating the fusion of background information and increasing the prominence of object features across low to high levels.

Quantitative Visualization: Fig. 7 displays the visualization of CALNet on the DroneVehicle, with accurate classification and localization of objects in dense small objects and low light conditions.

5 CONCLUSIONS

In this work, we propose CALNet, a novel approach for detecting multispectral objects that integrates cross-modal fusion and conflict rectification. Our CALNet method includes a CCR module that minimizes modal discrepancies by addressing semantically inconsistent multispectral features while leveraging cross-modal contextual information. Meanwhile, our SCF module ranks the intra-modal importance of features to select semantically rich ones and mines inter-modal complementary information to fuse multispectral features. We design a one-stage orientated detector based on CALNet to demonstrate the effectiveness of our approach in mitigating cross-modal semantic conflicts. Extensive experiments on two challenging multispectral object detection benchmarks validate that CALNet achieves state-of-the-art results and robustness.

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